

EP 3991102 A1 "Pruning and/or Quantizing Machine Learning Predictors" published 2022-05-04
US 2022/0114455 A1 "Pruning and/or Quantizing Machine Learning Predictors" published 2022-04-14
WO 2020/260656 A1 "Pruning and/or Quantizing Machine Learning Predictors" published 2020-12-30

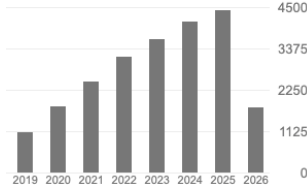
Relevance Score Assignment for Artificial Neural Networks 2016

US 20180018553 "Relevance Score Assignment for Artificial Neural Networks " granted 2024-10-04
CN 107636693 "Relevance Score Assignment for Artificial Neural Networks" granted 2022-01-11
EP 3271863 "Relevance Score Assignment for Artificial Neural Network" granted 2021-07-28
JP 6725547 "Relevance Score Assignment for Artificial Neural Networks" granted 2020-07-22
KR 102130162 "Assignment of Relevance Scores for Artificial Neural Networks" granted 2020-07-06
CA 2979579 "Relevance Score Assignment for Artificial Neural Networks" granted 2020-02-18
RU 2703343 "Relevancy Assessment for Artificial Neural Networks" granted 2019-10-16
BR 112017019821 "Relevance Score Assignment for Artificial Neural Networks " published 2018-05-15
WO 2016150472 A1 "Relevance score assignment for artificial neural network" published 2016-09-29

Publications

Summary of Scientific Impact

	All	Since 2021
# Publications	109	83
# Citations	23645	19630
h-index	40	39
i10-index	68	66



per Google Scholar, retrieved on May 30th, 2026.

List of Publications

Journal Articles

- Anders C J, Neumann D, Samek W, Müller K-R and **Lapuschkin S** (2026).
"Software for Dataset-wide XAI: From Local Explanations to Global Insights with Zennit, CoRelAy, and ViRelAy".
In: *PLoS ONE* 21(1):e0336683.
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- Cantú E D, Wittmann R K, Abdeen O, Wagner P, Samek W, Baier M and **Lapuschkin S** (2026).
"Deep Learning-based Multi Project InP Wafer Simulation towards Unsupervised Surface Defect Detection".
In: *Journal of the Franklin Institute* 363(2):108316
- Yolcu G Ü, Weckbecker M, Wiegand T, Samek W and **Lapuschkin S** (2025).
"Sparse, Efficient and Explainable Data Attribution with DualXDA".
In: *Transactions on Machine Learning Research* qfx81N884A.
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- Dreyer M, Berend J, Labarta T, Vielhaben J, Wiegand T, **Lapuschkin S** and Samek W (2025).
"Mechanistic Understanding and Validation of Large AI Models with SemanticLens".
In: *Nature Machine Intelligence* 1–14.
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- Pahde F, Wiegand T, **Lapuschkin S** and Samek W (2025).
"Ensuring Medical AI Safety: Explainable AI-Driven Detection and Mitigation of Spurious Model Behavior and Associated Data".
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6. Ma J, Weicken E, Pahde F, Weitz K, **Lapuschkin S**, Samek W and Wiegand T (2025).
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 In: *Bundesgesundheitsblatt – Gesundheitsforschung – Gesundheitsschutz* 68:915-923
7. Storås A M, Dreyer M, Pahde F, **Lapuschkin S**, Samek W, Halvorsen P, de Lange T, Mori Y, Hann A, Berzin T M, Parasa S and Riegler M A (2025).
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8. Weber L, Berend J, Weckbecker M, Binder A, Wiegand T, Samek W and **Lapuschkin S** (2025).
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9. Hedström A, Bommer P L, Burns T F, **Lapuschkin S**, Samek W and Höhne M-C M (2025).
 “Evaluating Interpretable Methods via Geometric Alignment of Functional Distortions”.
 In: *Transactions on Machine Learning Research* ukLxqA8zXj.
<https://github.com/annahedstroem/GEF/> | *TMLR Survey Certification*
10. Bley F, **Lapuschkin S**, Samek W and Montavon G (2025).
 “Explaining Predictive Uncertainty by Exposing Second-Order Effects”.
 In: *Pattern Recognition* 160:111171.
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11. Vielhaben J, **Lapuschkin S**, Montavon G and Samek W (2024).
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12. Becker S, Vielhaben J, Ackermann M, Müller K-R, **Lapuschkin S** and Samek W (2024).
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13. Achitibat R, Dreyer M, Eisenbraun I, Bosse S, Wiegand T, Samek W and **Lapuschkin S** (2023).
 “From attribution maps to human-understandable explanations through Concept Relevance Propagation”.
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14. Hedström A, Bommer P, Wickström K K, Samek W, **Lapuschkin S** and Höhne M-C M (2023).
 “The Meta-Evaluation Problem in Explainable AI: Identifying Reliable Estimators with MetaQuantus”.
 In: *Transactions on Machine Learning Research* j3FK00HyfU.
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15. Weber L, **Lapuschkin S**, Binder A and Samek W (2023).
 “Beyond Explaining: Opportunities and Challenges of XAI-Based Model Improvement”.
 In: *Information Fusion* 92:154–176
16. Hedström A, Weber L, Krakowczyk D G, Bareeva D, Motzkus F, Samek W, **Lapuschkin S** and Höhne M-C M (2023).
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17. Hofmann S M, Beyer F, **Lapuschkin S**, Golterman O, Loeffler M, Müller K-R, Villringer A, Samek W and Witte A V (2022).
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19. Rieckmann A, Dworzynski P, Arras L, **Lapuschkin S**, Samek W, Onyebuchi A A, Rod N H, Ekstrøm C T (2022).
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20. Slijepcevic D, Horst F, **Lapuschkin S**, Horsak B, Raberger A-M, Kranzl A, Samek W, Breiteneder C, Schöllhorn W I and Zeppelzauer M (2022).
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 In: *ACM Transactions on Computing for Healthcare* 3(2):14:1–27.
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21. Anders C J, Weber L, Neumann D, Samek W, Müller K-R and **Lapuschkin S** (2022).
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 In: *Information Fusion* 77:261–295
22. Sun J, **Lapuschkin S**, Samek W and Binder A (2022).
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 In: *Information Fusion* 77:233–246
23. Samek W, Montavon G, **Lapuschkin S**, Anders C J, and Müller K-R (2021).
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26. Horst F, Slijepcevic D, Zeppelzauer M, Raberger AM, **Lapuschkin S**, Samek W, Schöllhorn WI, Breiteneder C, and Horsak B (2020).
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27. Hägele M, Seegerer P, **Lapuschkin S**, Bockmayr M, Samek W, Klauschen F, Müller K-R and Binder A (2020).
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31. Montavon G, **Lapuschkin S**, Binder A, Samek W and Müller K-R (2017).
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Pattern Recognition Best Paper Award and Pattern Recognition Medal winner
32. Samek W, Binder A, Montavon G, **Lapuschkin S**, and Müller K-R (2017).
 “Evaluating the Visualization of what a Deep Neural Network has Learned”.
 In: *IEEE Transactions of Neural Networks and Learning Systems*

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In: *Journal of Neuroscience Methods* 274:141–145
34. **Lapuschkin S**, Binder A, Montavon G, Müller K-R and Samek W (2016).
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Contributions to Conference Proceedings and Workshops

1. Ostermann S, Gurgurov D, Baeumel T, Hedderich M A, **Lapuschkin S**, Samek W and Schmitt V (2026).
“From Weights to Activations: Is Steering the Next Frontier of Adaptation?”.
In: *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*
2. Komorowski P, Golimblevskaia E, Achibat R, Wiegand T, **Lapuschkin S** and Samek W (2026).
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3. Kahardipraja P, Achibat R, Wiegand T, Samek W and **Lapuschkin S** (2025).
“The Atlas of In-Context Learning: How Attention Heads Shape In-Context Retrieval Augmentation”.
In: *Advances in Neural Information Processing Systems (NeurIPS)* .
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4. Bareeva D, Höhne M M C, Warnecke A, Pirch L, Müller K-R, Rieck K, **Lapuschkin S** and Bykov K (2025).
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In: *Advances in Neural Information Processing Systems (NeurIPS)* .
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5. Sandmann E, **Lapuschkin S** and Samek W (2025).
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8. Puri B, Jain A, Golimblevskaia E, Kahardipraja P, Wiegand T, Samek W and **Lapuschkin S** (2025).
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9. Naujoks J, Krasowski A, Weckbecker M, Yolcu G Ü, Wiegand T, **Lapuschkin S**, Samek W and Klausen R P (2025).
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10. Erogullari E, **Lapuschkin S**, Samek W and Pahde F (2025).
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11. Joseph S, Suresh P, Hufe L, Stevinson E, Graham R, Vadi Y, Bzdok D, **Lapuschkin S**, Sharkey L and Richards A (2025).
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13. Bareeva D, Yolcu GÜ, Hedström A, Wiegand T, Samek W and **Lapuschkin S** (2024). "Quanda: An Interpretability Toolkit for Training Data Attribution Evaluation and Beyond". In: *NeurIPS 2024 Workshop on Attributing Model Behavior at Scale (ATTRIB 2024)* .
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Reproducibility Badge Winner
15. Kopf L, Bommer P L, Hedström A, **Lapuschkin S**, Höhne M M-C and Bykov K (2024). "CoSy: Evaluating Textual Explanations of Neurons". In: *Advances in Neural Information Processing Systems (NeurIPS)* 34656–34685. (OpenReview)
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18. Achitibat R, Hatefi S M V, Dreyer M, Jain A, Wiegand T, **Lapuschkin S**, Samek W (2024). "AttnLRP: Attention-Aware Layer-wise Relevance Propagation for Transformers". In: *Proceedings of the 41st International Conference on Machine Learning (ICML)* 135–168.
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19. Hatefi S M V, Dreyer M, Achitibat R, Wiegand T, Samek W and **Lapuschkin S** (2024). "Pruning By Explaining Revisited: Optimizing Attribution Methods to Prune CNNs and Transformers". In: *Proceedings of the European Conference on Computer Vision (ECCV) Workshops* 152–169. (Green Open Access)
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20. Hedström A, Weber L, **Lapuschkin S**, Höhne M M-C (2024). "A Fresh Look at Sanity Checks for Saliency Maps". In: *Proceedings of the 2nd XAI World Conference* 403–420. (Green Open Access)
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23. Bareeva D, Dreyer M, Pahde F, Samek W and **Lapuschkin S** (2024). "Reactive Model Correction: Mitigating Harm to Task-Relevant Features via Conditional Bias Suppression". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops* 3532–3541.
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- “Understanding the (Extra-)Ordinary: Validating Deep Model Decisions with Prototypical Concept-based Explanations”.
In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops* 3491–3501.
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 27. Frommholz A, Seipel F, **Lapuschkin S**, Samek W and Vielhaben J (2023).
“XAI-based Comparison of Audio Event Classifiers with different Input Representations”.
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 28. Hedström A, Weber L, **Lapuschkin S** and Höhne M M-C (2023).
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 29. Pahde F, Dreyer M, Samek W and **Lapuschkin S** (2023).
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 30. Binder A, Weber L, **Lapuschkin S**, Montavon G, Müller K-R and Samek W (2023).
“Shortcomings of Top-Down Randomization-Based Sanity Checks for Evaluations of Deep Neural Network Explanations”.
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 31. Dreyer M, Achtabat R, Wiegand T, Samek W and **Lapuschkin S** (2023).
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 32. Pahde F, Yolcu GÜ, Binder A, Samek W and **Lapuschkin S** (2023).
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 33. Krakowczyk D G, Prasse P, Reich D R, **Lapuschkin S**, Scheffer T, Jäger L A (2023).
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 34. Krakowczyk D G, Reich D R, Prasse P, **Lapuschkin S**, Jäger L A and Scheffer T (2022).
“Selection of XAI Methods Matters: Evaluation of Feature Attribution Methods for Oculomotoric Biometric Identification”.
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 35. Motzkus F, Weber L and **Lapuschkin S** (2022).
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 37. Sun J, **Lapuschkin S**, Samek W, Zhao Y, Cheung N-M and Binder A (2021).
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40. Sun J, **Lapuschkin S**, Samek W and Binder A (2020).
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42. Sun J, **Lapuschkin S**, Samek W, Zhao Y, Cheung N-M and Binder A (2020).
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In: *XXAI: Extending Explainable AI Beyond Deep Models and Classifiers. ICML Workshop*
43. Alber M, **Lapuschkin S**, Seegerer P, Hägele M, Schütt K T, Montavon G, Samek W, Müller K-R, Dähne S and Kindermans P-J (2018).
“How to iNNvestigate Neural Networks’ Predictors!”.
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